

MathQuestion-002

Review

$$\frac{ax^2}{a} + \frac{bx}{a} + \frac{c}{a} = \frac{0}{a}$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \quad \frac{b}{a} = p, \frac{c}{a} = q$$

$$x^2 + px + q = 0 \quad \text{Let } x = y + ?$$

$$(y + ?)^2 + p(y + ?) + q = 0 \leftarrow \text{don't want } y \text{ term}$$

$$(y + ?)^2 \rightarrow 2y \left[? = -\frac{p}{2} \right] \quad p(y + ?) \rightarrow py$$

$$x^2 + px + q = 0 \quad \text{Let } x = y - \frac{p}{2}, y = x + \frac{p}{2}$$

$$(y + ?)^2 + p(y + ?) + q = 0$$

$$y^2 - \cancel{py} + \frac{p^2}{4} - \cancel{py} - \frac{2 \cdot p^2}{2 \cdot 2} + q = 0$$

$$y^2 - \frac{p^2}{4} + q = 0$$

$$y^2 = \pm \sqrt{p^2 - q}$$

$$x + \frac{p}{2} = \pm \sqrt{\frac{p^2}{4} - q}$$

$$x = -\frac{p}{2} \pm \sqrt{\frac{p^2}{4} - q}$$

1 Question:

$$ax^3 + bx^2 + cx + d = 0$$

3 parts

1.get rid of x^2 term

2.solve $y^3 + py + q = 0$

3.put together

Part1

$$\text{let } x = y - \frac{b}{3a}$$

$$\begin{aligned} & a \left(y - \frac{b}{3a} \right)^3 + b \left(y - \frac{b}{3a} \right)^2 + c \left(y - \frac{b}{3a} \right) + d = 0 \\ & a \left(y^3 - 3y^2 \frac{b}{3a} + 3y \frac{b^2}{9a^2} - \frac{b^3}{27a^3} \right) + b \left(y^2 - 2y \frac{b}{3a} + \frac{b^2}{9a^2} \right) + cy - \frac{cb}{3a} + d = 0 \\ & ay^3 - \cancel{3ay^2 \frac{b}{3a}} + \cancel{3y \frac{b^2}{9a^2}} - \frac{b^3}{27a^3} + by^2 - \frac{2b^2}{3a}y + \frac{b^3}{9a^2} + cy - \frac{cb}{3a} + d = 0 \\ & ay^3 - \cancel{by^2} + \frac{b^2}{3a}y - \frac{b^3}{27a^2} - \cancel{by^2} - \frac{2b^2}{3a}y + \frac{3 \cdot b^3}{3 \cdot 9a^2} + cy - \frac{cb}{3a} + d = 0 \\ & \left(ay^3 + \left(\frac{-b^2}{3a} + c \right) y + \left(\frac{2b^3}{27a^3} - \frac{cd}{3a} + d \right) \right) \times \frac{1}{a} = 0 \times \frac{1}{a} \\ & y^3 + \left(\frac{-b^2}{3a^2} + \frac{c}{a} \right) (\rightarrow p) y + \left(\frac{2b^3}{27a^3} - \frac{cb}{3a^2} + \frac{d}{a} \right) \rightarrow q = 0 \end{aligned}$$

$$y^3 + py + q = 0$$

Part2

$$\begin{aligned}
 &\text{let } y = z + \frac{p}{z} \\
 &y^3 + py + q = 0 \\
 &\left(z + \frac{p}{z}\right)^3 + p\left(z + \frac{p}{z}\right) + q = 0 \\
 &z^3 + 3z^2 \cdot \frac{p}{z} + 3z \cdot \frac{p^2}{z^2} + \frac{p^3}{z^3} + pz + \frac{p}{z} + q = 0 \\
 &y = z - \frac{p}{3z}
 \end{aligned}$$

$$\begin{aligned}
 &y^3 + py + q = 0 \text{ let } y = z - \frac{p}{3z} \\
 &\left(z - \frac{p}{3z}\right)^3 + \left(z - \frac{p}{3z}\right) + q = 0 \\
 &z^3 - 3z^2 \cdot \frac{p}{3z} + 3z \cdot \frac{p^2}{9z^2} - \frac{p^3}{27z^3} + pz - \frac{p^2}{3z} + q = 0 \\
 &\left(z^3 - \frac{p^3}{27z^3} + q\right) \times z^3 = 0 \times z^3 \\
 &(z^3)^2 + q(z^3) - \frac{p^3}{27} = 0
 \end{aligned}$$

Earlier

$$\begin{aligned}
 &x^2 + Px + Q = 0 \\
 &x = -\frac{P}{2} \pm \sqrt{\frac{P^2}{4} - Q}
 \end{aligned}$$

$$z^3 = -\frac{q}{2} \pm \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}$$

$$\text{take } z = \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} \quad y = z - \frac{p}{3z}$$

$$y = \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} - \frac{p \cdot \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}}{3 \cdot \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} \cdot \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}}$$

$$y = \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} - \frac{p \cdot \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}}{3 \cdot \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} \cdot \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}}$$

$$y = \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} - \frac{p \cdot \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}}{\cancel{3} \cdot \sqrt[3]{\cancel{+\frac{q^2}{4}} - \frac{p^3}{\cancel{27}}}}}$$

$$y = \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}$$

$$\omega^3 = 1$$

$$\omega^3 - 1 = 0$$

$$(\omega - 1)(\omega^2 + \omega + 1) = 0$$

$$\omega = 1 \quad \omega = -\frac{1}{2} \pm \frac{i\sqrt{3}}{2}$$

$$\omega_0 = 1$$

$$\omega_1 = -\frac{1}{2} + \frac{i\sqrt{3}}{2}$$

$$\omega_2 = -\frac{1}{2} - \frac{i\sqrt{3}}{2}$$

$$y^3 + py + q = 0$$

$$y_1 = \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}$$

$$y_2 = \left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \left(-\frac{1}{2} - \frac{i\sqrt{3}}{2}\right) \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}$$

$$y_3 = \left(-\frac{1}{2} - \frac{i\sqrt{3}}{2}\right) \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}$$

Part3 Put together

$$\begin{aligned}
 x &= y - \frac{b}{3a} \quad y_1, y_2, y_3 \\
 p &= \frac{-b^2}{3a^2} + \frac{c}{a} \\
 q &= \frac{2b^3}{27a^3} - \frac{cb}{3a^2} + \frac{d}{a}
 \end{aligned}$$

$$ax^3 + bx^2 + cx + d = 0$$

$$\begin{aligned}
 x_1 &= \sqrt[3]{\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a} + \sqrt{\left(\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{-b^2}{9a^2} + \frac{c}{3a}\right)^3}} \\
 &\quad + \sqrt[3]{\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a} - \sqrt{\left(\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{-b^2}{9a^2} + \frac{c}{3a}\right)^3}} - \frac{b}{3a} \\
 x_2 &= \left(\frac{-1}{2} + \frac{i\sqrt{3}}{2}\right) \sqrt[3]{\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a} + \sqrt{\left(\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{-b^2}{9a^2} + \frac{c}{3a}\right)^3}} \\
 &\quad + \left(\frac{-1}{2} - \frac{i\sqrt{3}}{2}\right) \sqrt[3]{\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a} - \sqrt{\left(\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{-b^2}{9a^2} + \frac{c}{3a}\right)^3}} - \frac{b}{3a} \\
 x_3 &= \left(\frac{-1}{2} - \frac{i\sqrt{3}}{2}\right) \sqrt[3]{\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a} + \sqrt{\left(\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{-b^2}{9a^2} + \frac{c}{3a}\right)^3}} \\
 &\quad + \left(\frac{-1}{2} + \frac{i\sqrt{3}}{2}\right) \sqrt[3]{\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a} - \sqrt{\left(\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{-b^2}{9a^2} + \frac{c}{3a}\right)^3}} - \frac{b}{3a}
 \end{aligned}$$